

CREDIT RISK INTELLIGENCE

Credit Risk Intelligence Platform

Explainable risk decisions, not just a probability.

307,511

applicants scored

0.276

PR-AUC (ROC-AUC 0.782)

8.07%

base default rate

Business Context

A score a regulator can't question is a liability, not an asset.



Explainability

Fair-lending review and adverse-action notices require naming why an applicant was declined — not just a probability.



Business rules

Underwriters approve exceptions in if/then language, not model internals — rules translate the score into policy.

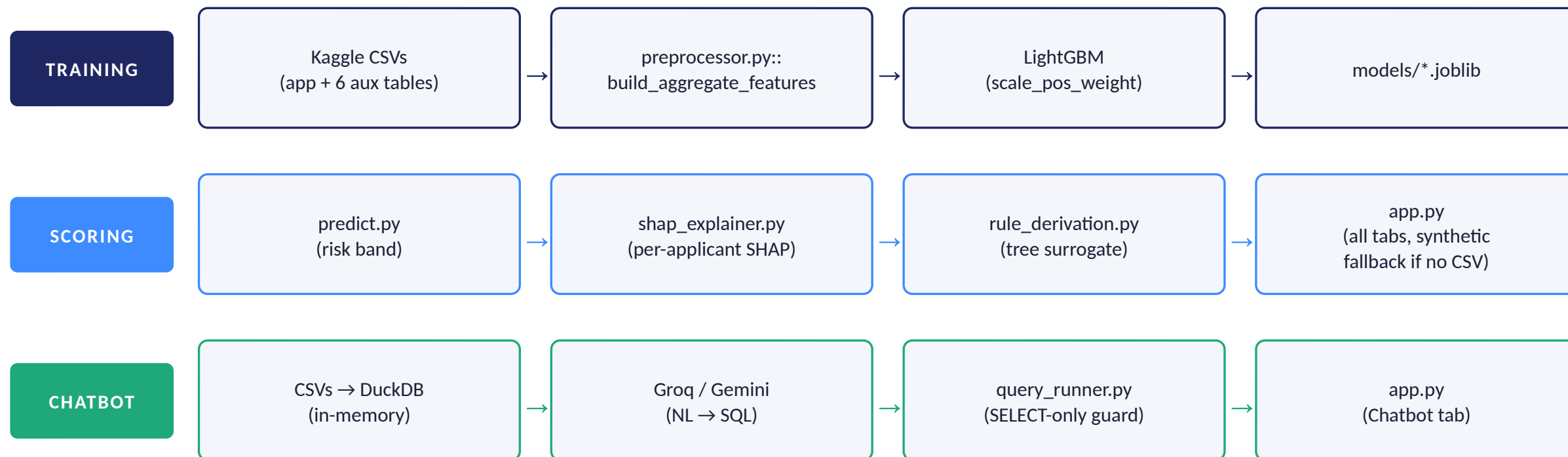


Talk-to-data

Risk and portfolio teams ask ad-hoc questions daily; a guarded natural-language interface answers without a SQL request queue.

Architecture

Three flows sharing one dataset and one Streamlit app



*Same aggregation function is shared by train.py, predict.py, and shap_explainer.py — inference-time features always match training-time features.
One container — docker compose up --build, volumes on ./data and ./models, config from .env.*

Chatbot has zero setup dependency: if application_train.csv is missing, query_runner.py auto-falls-back to a committed synthetic demo dataset (data/sample_applicants.csv) — every tab works immediately after docker-compose up --build, no Kaggle download required.

HOW TO RUN THIS

Docker-Based Evaluation

1

No dataset, no local Python — just a .env with a free Groq/Gemini API key, then docker-compose up --build.

2

EDA, Risk Prediction, Explainability, and Business Rules all run off the pretrained model already committed to the repo (models/credit_risk_model.joblib).

3

Chatbot runs off a synthetic 4,000-row demo dataset (also committed), with a clear on-screen banner: “ Using synthetic demo data — application_train.csv wasn't found...”

4

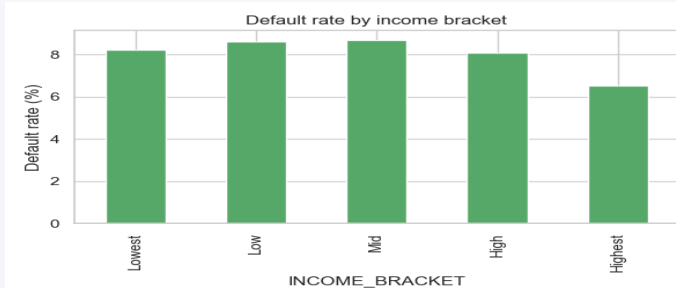
Placing the real Kaggle CSVs in data/ + docker-compose run --rm app python -m src.ml.train switches everything to real data automatically — no code changes, banner disappears.

5

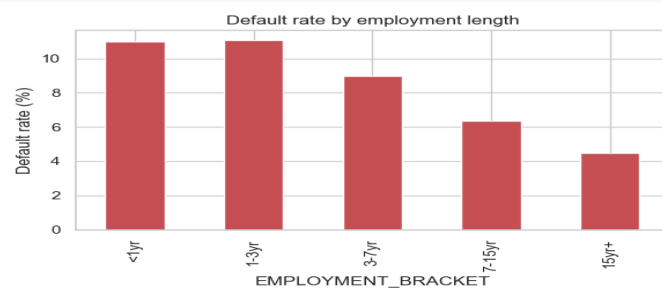
Chatbot can also serve real data on a deployed instance via a live Postgres attach (POSTGRES_URL) — no CSV, no re-download, dataset never touches git.

EDA Insights

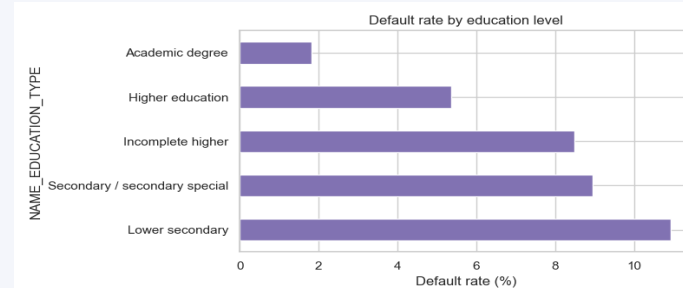
307,511 applicants • 8.07% default rate • six demographic/financial cuts



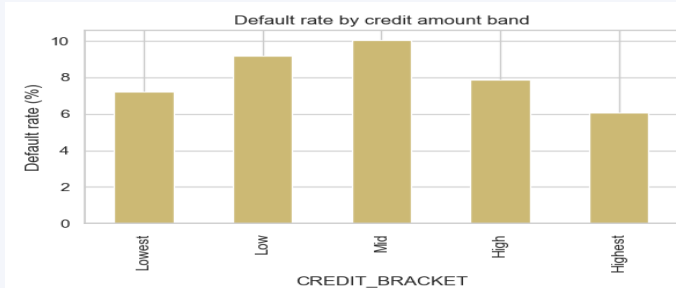
Default rate falls steadily as income bracket rises — lowest bracket defaults ~1.3× more than highest.



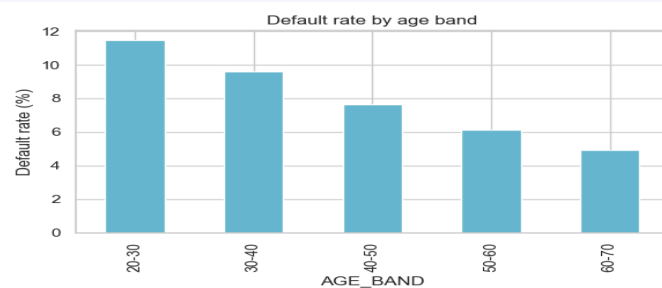
Shorter employment tenure → higher default rate — motivated the DAYS_EMPLOYED_RATIO feature.



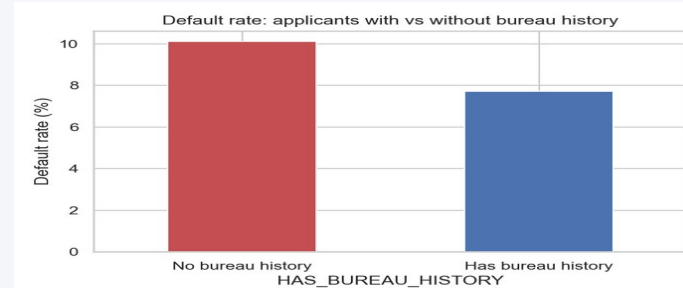
Academic-degree holders default at ~1/4 the rate of lower-secondary-educated applicants.



Default rate isn't monotonic in loan size — mid-sized credit amounts carry the highest risk.



Younger applicants default markedly more — the strongest single demographic split.



Applicants with no bureau history default more (10.12%) than those with history (7.73%) — a 2.39pp gap that's the empirical basis for joining bureau.csv, not just a domain-knowledge assumption.

Model & Results

Why LightGBM

Native missing-value handling, mixed numeric/categorical signal, strong out-of-box performance on ~300K rows × 130+ features.

Why scale_pos_weight over SMOTE

Documented failure mode: SMOTE's synthetic samples can land in unrealistic regions of a high-dimensional one-hot space. Didn't benchmark it here — went straight to the cheaper, standard first choice for tree-boosting. SMOTE remains available via --smote for comparison.

Why bureau / previous-app features

EDA insight #6 measured a real 2.39pp default-rate gap by bureau-history status — not assumed from general dataset knowledge.

Favors recall: catches ~70% of defaulters — appropriate for a screening tool, where a missed defaulter costs more than a false flag.

PR-AUC 0.276 (primary) • ROC-AUC 0.782 (secondary)

PR-AUC leads given ~92/8 class imbalance — it reflects performance on the minority (default) class better than ROC-AUC does.

Model	PR-AUC	ROC-AUC
LightGBM (deployed)	0.276	0.782
Logistic Regression (baseline)	0.245	0.765

- 5-fold CV: PR-AUC 0.268 ± 0.006 , ROC-AUC 0.779 ± 0.004 — confirms the single split wasn't a lucky draw.
- Monotonic constraints (income, EXT_SOURCE_1/2/3 — higher always means lower risk) cost essentially nothing (-0.0001 PR-AUC) while making the model defensible to a fair-lending review.
- Bounded hyperparameter search found only $+0.002$ ROC-AUC — smaller than CV noise, so defaults were kept.

Confusion matrix @ 0.5 (LightGBM, with bureau features)

	Pred: No Default	Pred: Default
Actual: No Default	41,272	15,266
Actual: Default	1,516	3,449

Explainability (SHAP)

SHAP TreeExplainer runs at inference time, translated into plain language.

Sample applicant — scored 83.4% (High risk)

EXT_SOURCE_3 = 0.139	increased risk		0.72
EXT_SOURCE_1 = 0.083	increased risk		0.55
EXT_SOURCE_2 = 0.263	increased risk		0.38
DAYS_BIRTH = -9,461	decreased risk		0.30
DEF_60_CNT_SOCIAL_CIRCLE = 2	increased risk		0.13

For this applicant, the three EXT_SOURCE fields are the strongest contributors to the SHAP explanation.

Model Evaluation Tab

Live in the app, not just the README — reconstructs train.py's exact validation split.

[Screenshot placeholder — ROC curve · PR curve · confusion matrix heatmap]

Replace with a screenshot of the app's Model Evaluation tab

Business Rules

A shallow depth-4 decision tree trained as a surrogate on the model's own predictions.

HIGH RISK

```
IF EXT_SOURCE_3 ≤ 0.55 AND EXT_SOURCE_2 ≤ 0.52  
  AND EXT_SOURCE_1 ≤ 0.62 AND EXT_SOURCE_2 ≤  
  0.35  
THEN High Risk (likely default)
```

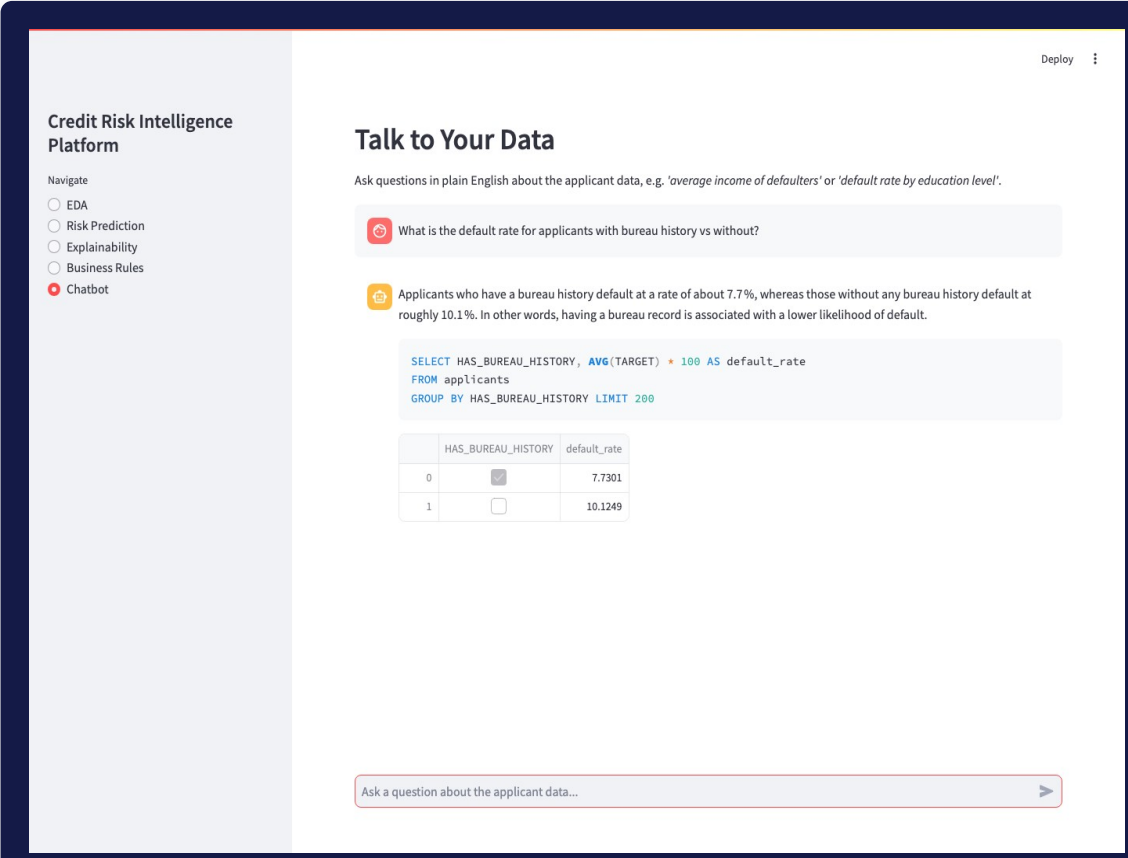
LOW RISK

```
IF EXT_SOURCE_3 > 0.55 AND EXT_SOURCE_2 ≤ 0.42  
  AND EXT_SOURCE_2 > 0.15 AND DAYS_EMPLOYED ≤ -  
  1,624.5  
THEN Low Risk (likely repay)
```

16 rules derived per run, surfaced live in the app's Business Rules tab.

Chatbot Demo

- Groq (openai/gpt-oss-20b, free) turns the question into SQL; Gemini free tier is the fallback.
- Every query validated via sqlparse: rejects non-SELECT statements, rejects unknown tables/columns.
- Retries once on a validation error before failing gracefully.
- Row-limit enforced on every query.
- 3-turn conversation memory for natural follow-ups.
- Schema now also exposes the new engineered fields — BUREAU_COUNT, HAS_BUREAU_HISTORY, PREV_APP_APPROVAL_RATE — so the chatbot can answer questions about credit history, not just the original application fields.
- Degrades gracefully with no dataset present — shown right is the synthetic-fallback banner, which appears automatically and is directionally honest (bureau-history correlation matches the real EDA finding).



Credit Risk Intelligence Platform

Navigate

- ☐ EDA
- ☐ Risk Prediction
- ☐ Explainability
- ☐ Business Rules
- ☒ Chatbot

Talk to Your Data

Ask questions in plain English about the applicant data, e.g. 'average income of defaulters' or 'default rate by education level'.

What is the default rate for applicants with bureau history vs without?

Applicants who have a bureau history default at a rate of about 7.7%, whereas those without any bureau history default at roughly 10.1%. In other words, having a bureau record is associated with a lower likelihood of default.

```
SELECT HAS_BUREAU_HISTORY, AVG(TARGET) * 100 AS default_rate
FROM applicants
GROUP BY HAS_BUREAU_HISTORY LIMIT 200
```

	HAS_BUREAU_HISTORY	default_rate
0	☒	7.7301
1	☐	10.1249

Ask a question about the applicant data...

Live app screenshot — Chatbot tab, querying the new bureau-history field

Limitations & Next Steps

- 1 Surrogate-tree rules can disagree with the model near split edges — explanatory aid, not a replacement scoring engine.
- 2 EXT_SOURCE_* and age are top predictors — need fair-lending / disparate-impact review before production use.
- 3 SQL guardrail is a token-scan via sqlparse, not a full parser.
- 4 All 6 auxiliary tables are now joined (previously only bureau + previous_application) — next layer is recency-weighted aggregates (last 6 months vs. all-time) and second-order trend features.
- 5 The +0.008 ROC-AUC lift is real but modest — validate on a time-based split before relying on it for risk-appetite decisions, since a random split may not test temporal drift.
- 6 Bounded hyperparameter search run — found only marginal gains, kept defaults.
- 7 Synthetic fallback data is calibrated only to the documented base rate (8%) and the bureau-history default-rate gap — a demo convenience, not a substitute for the real dataset in any evaluation of model quality.
- 8 Monotonic constraints currently cover only 4 of 291 features — a deliberately conservative starting set, not an exhaustive fair-lending review.